



अश्वाMOBILITY

LIVE IT. LOVE IT. RACE IT.

SIMULATION
TO SPEED.

ASHWA RACING

**JULY
2025
EDITION**

CONTENTS

SL NO.	TITLE	PAGE NO.
1	VISION AND MISSION	3
2	FROM CAD TO TRACK: THE DESIGN REVIEW.	5
3	ALUMNI CORNER	7
4	THROTTLE TALK	10
5	OUR SPONSORS	13
6	TEAM DIRECTORY	16

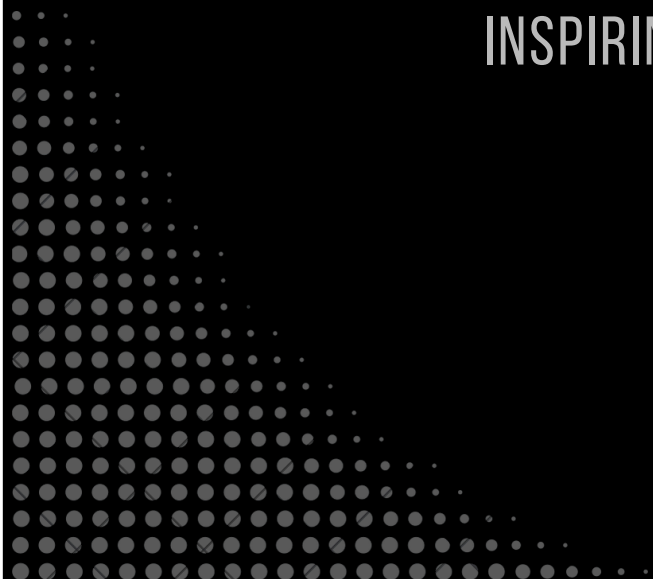


“

”

**VISION
AND
MISSION**

GUIDING PURPOSE,
INSPIRING FUTURE.



> MISSION AND VISION <

VISION

- To be the national leader in all categories of Formula Student, to push the limits of engineering and compete in events at the highest level.
- To foster student mobility solutions in the future by developing sustainable prototypes across various domains.

MISSION

- Prepare students for the industry by designing and building world-class vehicles while facilitating industry interaction.
- Showcase India's engineering expertise in mobility on a global stage and provide talent opportunities for the industry.
- Develop cost-effective, sustainable transportation solutions for the masses.

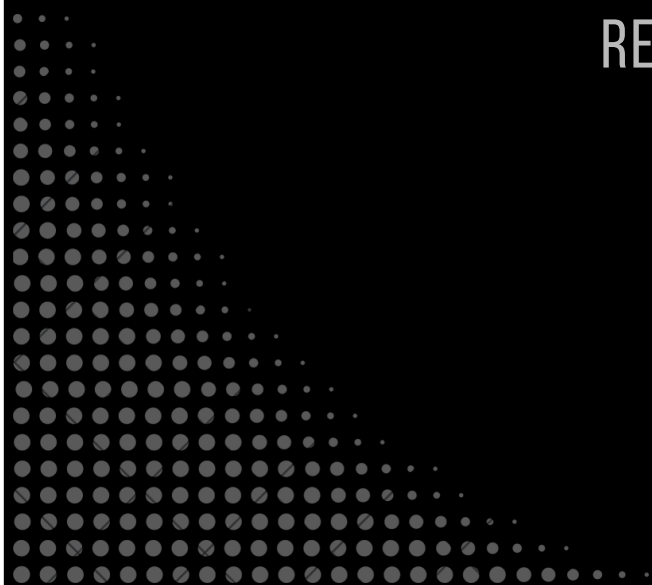


“

”

**FROM
CAD TO
TRACK**

THE DESIGN
REVIEW.





FROM CAD TO TRACK: THE DESIGN REVIEW.

The first design review of the season was successfully conducted, marking a major milestone in our project timeline. Each subsystem—Drivetrain, Suspension, Chassis, Engine, Brakes, and Electrical—presented their design progress, covering key areas such as material selection, calculations and simulations, CAD models, sub-assembly development, rule compliance, and considerations for manufacturability and serviceability.

The review followed a detailed evaluation rubric, with each criterion assessed on a scale from Unacceptable to Excellent. This structured format ensured consistency and helped highlight both strengths and areas needing improvement.

All team leads and subsystem members participated actively, and valuable feedback was collected throughout the day. Within 24 hours, the Chief Engineer categorized the feedback into three actionable levels—Must Fix, Need to Fix, and Nice to Have—setting the stage for the next phase of refinement. The review not only helped us assess technical readiness but also brought the entire team one step closer to a cohesive and production-ready design.



Fig. Subsystems presenting design models and progress during the design review.

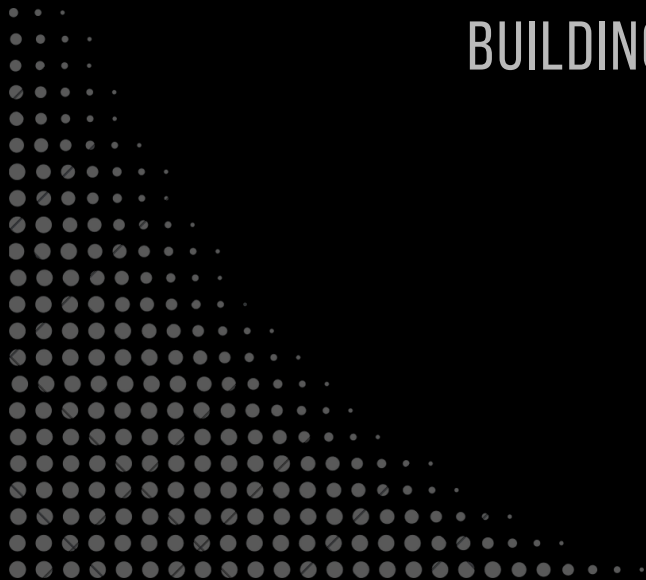


“

”

ALUMNI CORNER

CONNECTING MEMORIES,
BUILDING FUTURES!



ALUMNI CORNER



Manasi Kadambi, Partnerships, Marketing and PR, 2020

My journey with Ashwa Racing:

My journey with Ashwa Racing began in 2017 when I joined the Sponsorship, Marketing & PR team. As a Biotechnology major, it was a bold step into unfamiliar territory—but one that quickly became a defining chapter in my undergrad life. Over the next three years, I worked on everything from reaching out to potential sponsors and closing deals, to showcasing at high-visibility events like IMTEX and managing our brand presence. It wasn't just about raising funds—it was about telling our story, creating value, and ensuring the team had what it needed to build the car we all believed in. Ashwa pushed me beyond my comfort zone and sparked my love for tech, strategy, and the power of great execution.

My competition experience:

As part of the 2020 team, I had the chance to compete at two incredible events—Formula Hybrid+Electric 2019 in the U.S. and Formula Bharat 2020. For FH+E, I led sponsorship efforts and experienced my first international engineering competition. It gave me perspective—not just on tech, but on how teams across the world think, innovate, and collaborate. At Formula Bharat, I delivered our Business Plan Presentation and we placed 3rd. That experience, both in preparation and in the pitch room, shaped how I think about product, value propositions, and storytelling even today.

How Ashwa influenced my growth as an engineer:

Ashwa taught me that great engineering and innovation don't exist in silos—they need structure, planning, and strong cross-functional communication to make an actual impact. I learned early on that solving real-world problems is not just about building the right solution, it's about identifying the right problem, aligning with stakeholders, and making sure your solution makes business sense. That foundational understanding is what led me to my first role in product marketing at a fintech firm, and eventually shaped my path toward product management.

Standout projects or learnings:

Working on the Business Plan Presentation for Formula Bharat was a game-changer. It brought together everything I loved: strategy, creativity, business thinking, and clear communication. That single project played a major role in landing my first job and set the tone for everything that followed. I went on to pursue my Master's in Engineering Management at Northeastern University with a specialization in product management and tech innovation. I did my co-op with Honeywell Intelligrated in 2023 and have now joined their Offering Management team full-time. From biotech to fintech to industrial automation—it's been a winding road, but what's helped me stay grounded is the ability to learn fast, solve problems deeply, and adapt my skills across industries.

How the team shaped my personal and professional development:

Ashwa was a crash course in teamwork, resilience, and growth. It was the first time I worked so closely with people from different disciplines, all united by the shared mission of building the car. We had high highs and tough lows (FB 2020 didn't go as planned), but I learned that failure isn't the end of the story. What matters more is how you reflect, realign, and keep moving forward. Being one of the very few women on the team, I also understood early on that showing up mattered—being confident, competent, and consistent was key. It wasn't about fighting the status quo, it was about showing what's possible when you rise above it.

Closing note:

The Formula Student community has given me more than I could've imagined, and it's something I've been proud to give back to. I've served as a judge for the Business Plan Presentation at Formula Bharat since 2021 (now leading the event), and at Formula Hybrid+Electric from 2022 to 2024. Every time I interact with young, driven teams, I'm reminded of what Ashwa gave me—clarity, direction, and a tribe of people who want to build, challenge, and grow. For that, I'll always be grateful.



“

”

THROTTLE TALK

DECODING THE ELECTRONIC
THROTTLE SYSTEM





THROTTLE TALK: DECODING THE ELECTRONIC THROTTLE SYSTEM

Ever wondered how your bike's quickshifter works? Or how switching between rain, sport, and eco modes completely transforms the feel of your ride? Or how traction control steps in to save countless riders from a crash? All of this is made possible thanks to advancements in the Electronic Throttle Body (ETB) system.

At Ashwa Racing, we have been using the Electronic Throttle Body (ETB) system since 2024, enabling the implementation of advanced features such as quickshifter and auto-blipper.



Fig. The accelerator pedal with a 6 pin connector for the apps housing two rotary position sensors.

The primary components of the Electronic Throttle System include the Accelerator Pedal Position Sensor (APPS) and the Electronic Throttle Body itself, which contains the butterfly valve responsible for regulating the airflow into the engine.

We use an APPS from Williams Controls, which is attached to the accelerator pedal housing. It consists of two inductive rotary position sensors along with a signal conditioning circuit that outputs the positions of both independent sensors. These signals are sent to the MoTeC M800 ECU, which then calculates the target position of the butterfly valve in the ETB.

The MoTeC ECU sends a PWM signal to the ETB's stepper motor, which actuates the butterfly valve accordingly. The ETB also includes two Throttle Position Sensors (TPS) that continuously measure the actual position of the valve and feed this information back to the ECU for closed-loop control.

Now that the basics of the Electronic Throttle System are out of the way, here comes the fun part! Using the MoTeC M800 ECU, we can detect when the upshift button is pressed and momentarily cut power by interrupting ignition or fuel for a short duration. This allows seamless clutchless upshifting on our sequential gearbox.

When the downshift button is pressed, the ECU uses the ETB to automatically blip the throttle, momentarily increasing engine RPM to match the gearbox input speed. This rev-matching technique allows smooth, clutchless downshifting.

The Electronic Throttle Body (ETB) also enables precise and stable idle control through a closed-loop PID (Proportional-Integral-Derivative) algorithm. By allowing fine-tuning of each term—proportional, integral, and derivative—within the MoTeC M800 ECU, we can accurately manage the throttle opening to maintain a consistent idle speed under varying engine loads and conditions. The ETB's ability to make real-time, ECU-controlled adjustments eliminates the need for mechanical idle valves, simplifying the system while improving its responsiveness and reliability.

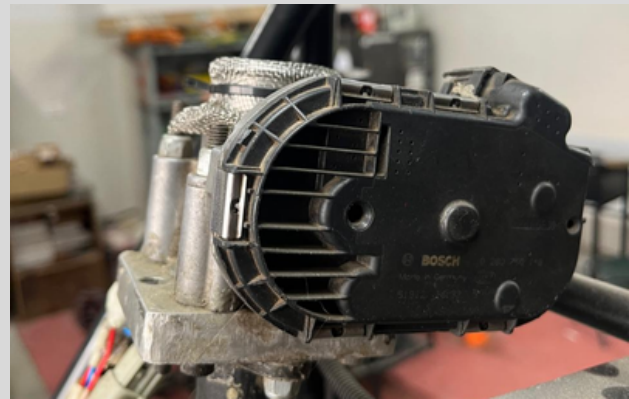


Fig. The Bosch ETB housing the butterfly valve and the stepper motor.

Safety, Served by Wire

To ensure the safety of the ETB system, multiple layers of hardware and software checks are implemented, starting with the use of redundant sensors. Both the Accelerator Pedal Position Sensor (APPS) and the Throttle Position Sensor (TPS) are equipped with dual independent channels. This redundancy allows the ECU to constantly verify the plausibility of the signals by comparing the two channels for consistency.

In the case of the APPS, we use a dedicated APPS board that monitors both channels in real time. If the two signals differ by more than 10%, the board flags this as a fault and triggers a safety mechanism—cutting power to the ETB to prevent any unintended throttle movement. A similar check is done for the TPS, comparing the feedback signals from the throttle body to ensure that the actual throttle position matches the commanded position within the same 10% tolerance.

Additionally, since the APPS and TPS signals are Single Channel Sensor (SCS) types, the system includes short-circuit checks. On the XX5C which was the 2025 prototype, an arduino nano is used to monitor the signal lines for unexpected voltage levels that could indicate a short to ground or supply. If a fault is detected, these circuits immediately trigger the Shutdown Circuit (SDC) which cuts power to the fuel injector, fuel pump and ignition.

Together, these checks form a robust fail-safe architecture that allows the ETB system to operate safely and reliably even in the demanding conditions of a race environment.

These integrated checks create a strong, fail-safe architecture, enabling the ETB system to function safely and dependably, even under the rigorous demands of a race environment.

*Akshay Urs,
Electrical Member,
'27 Batch Ashwa Racing.*



“

”

**OUR
SPONSORS
(2025-2026)**

OUR SPONSORS BRIDGE INDUSTRY AND ACADEMIA,
DRIVING US TO GLORY!



OUR SPONSORS



SINEWAVE GENERATORS
PRIVATE LIMITED



DYNAMATIC
TECHNOLOGIES



MERITOR®





OUR SPONSORS



*Joe's
Garage™*



DMG MORI



wilwood DISC BRAKES

DS **SOLIDWORKS**



Ansys



AKSHAYA MOTORS





“

”

**TEAM
DIRECTORY
2026**

DRIVEN BY PASSION, UNITED BY ENGINEERING—
THIS IS ASHWA RACING!

FACULTY ADVISOR

Dr. Ravindra S. Kulkarni

FOUNDER AND COACH

Arunraj Subbaraj

2026 BATCH

DRIVETRAIN AND HR

Adithya Ranjith	+91 76196 94442
Amitha A.V. (Team Captain)	+91 96063 79252
Kiran Kumar M	+91 6363 484 776

ELECTRICAL AND TESTING

Dhruva S Kashyap	+91 6364 427 004
G Venkat Shreyas	+91 99726 70533
Sandeep N Uttarkar	+91 93802 18325
Sudharsan S	+91 98445 77820
Vaibhav MB	+91 8088412636
Vibin S	+91 72043 69125
Viraj Bhat B	+91 9535468677

SUSPENSION AND ADMIN

Kritik Modawel (Project Manager)	+91 96323 96324
Kunal Dileep Bhairodgi	+91 97855 48544
Nischitha DV	+91 83104 67458
Yashas Bharamagoudar	+91 94812 12403
Amish Srivastava	+91 9620182043

CHASSIS AND WORKSHOP MANAGEMENT

Amulya K.V.	+91 87626 86499
Bhuvan Krishna Reddy P	+91 97035 70555
Gaurav Raju (Chief Engineer)	+91 72592 54288
Hanish Varma Manthena	+91 93913 48565
Karthik S	+91 97312 37861

INFORMATION TECHNOLOGY

Aditya Saiprasad	+91 91081 26956
------------------	-----------------

SPONSORSHIP, MARKETING & PR

Syed Nadeem	+91 73380 79545
UP Madhavan	+91 9741085287
Vansh Vikas Jain	+91 82080 72332

BRAKES AND LOGISTICS

Hemanth BS	+91 76765 85140
------------	-----------------

ENGINE

Raguraam S	+91 99868 23451
------------	-----------------

2027 BATCH

DRIVETRAIN AND HR

Ranjith DS	+91 99004 53526
Srividya Sriraman	+91 82174 95430
Lok Rajan PN	+91 63607 01056
Nikhil Kashyap	+91 86602 39216
Rohith PS	+91 93534 32169
Aman Kumar Gupta	+91 91294 53406

ELECTRICAL AND TESTING

Aashrith A Rao	+91 99161 38855
Aditeya Sarkar	+91 63661 24242
Ayush Hegde	+91 89706 73624
Eshwar Prasad M	+91 94822 58710
G.Nandakishore Shankar Bharadwaj	+91 91486 73282
Krish Naveen	+91 99723 77610
N Akshay Urs	+91 96065 00196
Pratham.M.Rao	+91 63642 19485
Rohan Jagadeesha Reddy	+91 96325 75722
S Prajan	+91 91088 19393
Vinith Prabhu	+91 99000 43985
Tejas S Math	+91 93531 49945

CHASSIS AND WORKSHOP MANAGEMENT

Achinthya G Kadekar	+91 63660 95611
Amogh P Giri	+91 96320 59883
Krishna Rastogi	+91 63530 06119
Harshavardhan Yale	+91 70198 39583
Sathvik Sukeeth Naik	+91 70223 57439
Prasanna J Naik	+91 93800 92279

BRAKES AND LOGISTICS

Preetham P	+91 96639 99841
Shravan C S	+91 90195 35410
Rohan S	+91 96863 48424
Harshavardhan Premkumar	+91 63616 69525

SUSPENSION AND ADMIN

Aitijhya Sahoo	+91 87773 80853
Nihal Sathish	+91 98453 44007
Anshul Joshi	+91 72250 95562
Smriti J Abraham	+91 99725 35387
Yash Prashant Mistry	+91 81399 82001

ENGINE

K M Talin Thimmaiah	+91 80883 76612
Tejas	+91 78924 67274
Uday	+91 81478 78800
Bhuvan S.U	+91 86605 34024

SPONSORSHIP, MARKETING & PR

Samith Belegadde Sirish

+91 93533 43650

FINANCE

Siddhant Shah

+91 91047 81380

Pranav Ramakrishnan

+91 91487 18718

INFORMATION TECHNOLOGY

Arya Shetty

+91 72250 95562

Prajwal TA

+91 73384 29315

Prajwal TS

+91 82173 92640

Contact us:

- **Sponsorship, Marketing and PR:** ashwasmp@gmail.com
- **Suspension and Admin:** arf.admn@gmail.com
- **Finance:** arf.finance@gmail.com
- **Brakes and Logistics:** arf.infralogscm@gmail.com
- **HR:** ashwa.hr@gmail.com
- **Electrical and Testing:** arf.testing@gmail.com
- **Chassis and Workshop Management:**
arf.workshop@gmail.com
- **Engine:** arf.engine@gmail.com
- **IT:** arf.itnetwork@gmail.com